1. Создала каталог расширения
   mkdir -p ~/postgres/extensions/demo_extension
2. .control файл

```
comment = 'Demo extension: hello_world'
default_version = '1.0'
module_pathname = '$libdir/demo_extension'
relocatable = true
```

3. SQL-файл

```
/* contrib/dict_int/dict_int--1.0.sql */

-- complain if script is sourced in psql, rather than via CREATE EXTENSION
\echo Use "CREATE EXTENSION demo_extension" to load this file. \quit

CREATE FUNCTION hello_world()
        RETURNS text
        AS 'MODULE_PATHNAME', 'hello_world'
        LANGUAGE C IMMUTABLE;
```

4. .c файл

```
#include "postgres.h"
#include "fmgr.h"
#include "utils/builtins.h"

PG_MODULE_MAGIC;
PG_FUNCTION_INFO_V1(hello_world);

Datum
hello_world(PG_FUNCTION_ARGS)
{
    ereport(NOTICE,
        (errmsg("Hello, World from extension!")));
    PG_RETURN_TEXT_P(cstring_to_text("Hello, World from extension!"));
}
```

5. Makefile

```
MODULES = demo_extension
EXTENSION = demo_extension
DATA = demo_extension--1.0.sql
PGFILEDESC = "demo_extension - my hello_world() extension"

ifdef USE_PGXS
PG_CONFIG = pg_config
PGXS := $(shell $(PG_CONFIG) --pgxs)
include $(PGXS)
endif
```

6. Сборка через PGXS
   make USE_PGXS=1
   После успешной сборки проверила, что файлы появились ls -la
   total 96
   drwxr-xr-x 9 user staff 288 Oct 7 19:08 .
   drwxr-xr-x 3 user staff 96 Oct 3 22:19 ..
   drwxr-xr-x@ 3 user staff 96 Oct 5 19:08 .vscode
   -rw-r--r--@ 1 user staff 311 Oct 7 18:24 demo_extension--1.0.sql
   -rw-r--r--@ 1 user staff 304 Oct 7 19:08 demo_extension.c
   -rw-r--r--@ 1 user staff 125 Oct 4 11:24 demo_extension.control
   -rwxr-xr-x@ 1 user staff 9360 Oct 7 19:08 demo_extension.dylib
   -rw-r--r--@ 1 user staff 17880 Oct 7 19:08 demo_extension.o
   -rw-r--r--@ 1 user staff 238 Oct 7 19:05 Makefile
7. Установка
   make USE_PGXS=1 install
8. Запуск сервера
   pg_ctl -D ~/postgres/data/debug -l ~/postgres/log/debug/postgres.log start
9. Подключилась к базе
   psql -d postgres
10. Проверила, что расширение видно серверу
    SELECT * FROM pg_available_extensions WHERE name = 'demo_extension';

```
postgres=# SELECT * FROM pg_available_extensions WHERE name = 'demo_extension';
      name      | default_version | installed_version |           comment
----------------+-----------------+-------------------+-----------------------------
 demo_extension | 1.0             |                   | Demo extension: hello_world
(1 row)
```

11. Создала расширение в текущей базе
    CREATE EXTENSION demo_extension;

12. Вызов функции

    SELECT hello_world();

```
postgres=# CREATE EXTENSION demo_extension;
CREATE EXTENSION
postgres=# SELECT hello_world();
NOTICE:  Hello, World from extension!
         hello_world
-------------------------------------
 Hello, World from extension!
(1 row)
```

13. Проект на гитхаб https://github.com/cherkaasova/PostgresPro